PERFORMANCE WORK STATEMENT FOR IDIQ CONTRACT
ENVIRONMENTAL LABORATORY SERVICES
U.S. ARMY CORPS OF ENGINEERS, CHICAGO DISTRICT

February 24, 2026

1 PURPOSE

The purpose of this document is to procure an environmental sample analysis contract in support of the U.S. Army Corps of Engineers (USACE) Chicago District's Civil Works and Operations and Maintenance (O&M) missions. The services provided by this analytical contract may be utilized by other Districts within the boundaries of the Great Lakes and Ohio River Division (LRD) in support of their respective missions that require analytical services. The contract will be a 5-year Indefinite Delivery Indefinite Quantity (IDIQ) award in which task orders are written against the base contract.

2 GENERAL

This is a non-personal service(s) contract under which the personnel rendering the service(s) are not subject, either by the contract's terms or by the manner of its administration, to the supervision and control usually prevailing in relationships between the Government and its employees.

2.1 Description of Services / Introduction

This Performance Work Statement (PWS) describes the analytical services support that will be provided to USACE Chicago District and other Districts within the Great Lakes and Ohio River Division (LRD), required for the characterization of soils, sediments, ground waters, surface waters, tissues, oils, wastes, and other environmental sample types. Typical projects will involve analyzing samples collected to determine the concentrations of various pollutants and/or physical properties of collected media. The USACE is concerned about pollutants at trace concentration levels. See Table 1 for some expected chemical concentration Reporting Limit ranges and associated units. Concentrations of interest include the following: 1) typical of ambient open water, 2) low-levels required for comparison to risk-based screening criteria for water, soil and/or sediment quality, and 3) waste characterization screening. In addition, samples may be required to undergo several predictive tests including leaching tests, elutriate tests, and other laboratory prepared extractions.

Table 1. Anticipated Reporting Limit Ranges for Common Chemicals of Concern

Parameter	Reporting Limit Range	
	Solid (mg/kg)	*Aqueous (mg/L)*
Aluminum	1.0 -- 10.0	--
Arsenic	0.05 -- 1.0	0.005 -- 0.05
Barium	1.0 -- 5.0	0.01 -- 0.05
Cadmium	0.001 -- 0.1	0.001 -- 0.005
Chromium	0.01 -- 0.05	0.01 -- 0.05
Copper	0.005 -- 0.5	0.005 -- 0.025
Cyanide	0.1 -- 0.5	0.01 -- 0.05
Iron	0.5 -- 10.0	0.1 -- 0.5
Lead	0.05 -- 1.0	0.01 -- 0.05
Manganese	0.5 -- 1.0	0.1 -- 0.5
Mercury (Low Level)	0.01 -- 0.02	0.000001 -- 0.0002
Nickel	0.5 -- 5.0	0.005 -- 0.01
Selenium	0.1 -- 0.3	0.005 -- 0.01
Silver	0.05 -- 1.0	0.001 -- 0.003
Zinc	0.5 -- 2.0	0.02 -- 0.05
Total Phosphorus	0.001 -- 1.0	0.002 -- 0.02
Ammonia Nitrogen	0.02 -- 1.0	0.001 -- 0.01
Total Kjeldahl Nitrogen (TKN)	0.5 -- 1.0	--
Oil & Grease	10.0 -- 50.0	1.0 -- 5.0
Chemical Oxygen Demand	5.0 -- 10.0	5.0 -- 10.0
Hardness	--	1.0 -- 2.0
Sulfate	--	5.0 -- 10.0
Chloride	--	1.0 -- 2.0
Total Organic Carbon	50.0 -- 100.0	--
Alkalinity	--	10.0 -- 20.0
Total PCBs	0.01 -- 0.05	0.001 -- 0.05
TDS	--	1.0 -- 10.0
TSS	--	1.0 -- 10.0
PAHs	0.003 -- 0.01	0.001 - 0.1
VOCs	0.01 -- 0.05	0.001 -- 0.005
Pesticides	0.001 -- 0.03	--
TVS	1.0 -- 5.0 %	--

2.2 Background

The USACE Civil Works and Operations and Maintenance (O&M) missions that necessitate this Analytical Services Contract are designated as programs that focus on water resource development, navigation, and environmental stewardship. Site characterization via analytical services performed on sampled environmental media provides projects under these missions the ability to plan long-term dredging strategies, inform construction projects, and determine the environmental impacts of water control features through water quality monitoring.

2.3 Scope

The Contractor shall provide non-personal service(s). The work to be done hereunder consists of furnishing and delivering to the USACE all supplies and services necessary to conduct laboratory services as requested by specific task orders. The Contractor shall furnish all personnel, equipment, and supplies needed to provide analytical laboratory services for projects within the boundaries of the Great Lakes and Ohio River Division (LRD), USACE. Contractor’s performance will be in accordance with this PWS. A “typical” analysis project will require the analysis of sediment and water samples collected from harbors, their connecting channels, and inland waterways within the boundaries of the USACE Chicago District (Illinois, Indiana, Wisconsin) and will occasionally require work from other Districts within the Great Lakes and Ohio River Division.

Any work performed under this contract that is related to analysis of Hazardous, Toxic and Radiological Waste (HTRW), identified in individual task orders, shall meet the applicable portions of the Department of Defense Quality Systems Manual (QSM) located here:

https://www.denix.osd.mil/edqw/documents/documents/qsm-version-5-1-final/.

Since litigation over possible environmental effects may arise, it is essential that laboratory tests ensure that the production of data is to the highest degree of confidence, reproducibility, and reliability achievable by approved state-of-the-art and current methodologies. Full control over sample integrity and/or loss of sample constituents during the total process of sampling, sample handling and preservation, and testing, is necessary to obtain the quality of results the USACE requires. Pursuant to this, analysis must follow the approved procedures, quality assurance, and control requirements described in the following discussion of specifications for analysis services.

Chemical analyses required may include any or all parameters regulated under the Safe Drinking Water Act, the Clean Water Act, the Resources Conservation and Recovery Act, or Superfund program. Nonmetallic analyses routinely consist of chemical oxygen demand (COD), ammonia-nitrogen, total Kjeldahl nitrogen, total phosphorus, total residue, total volatile residue, total phenolics, total cyanide, total organic carbon (TOC), particle size distribution, and pH. Analyses for organic compounds may include persistent chlorinated hydrocarbons such as polychlorinated biphenyls (PCBs), pesticides, herbicides, various volatile and semi-volatile organic compounds (VOCs and SVOCs), including polycyclic aromatic hydrocarbons (PAHs), and emerging contaminants of concern. Water samples collected in support of the District's water quality mission include analysis of samples for phytoplankton and zooplankton species, chlorophyll ABC, and macroinvertebrates. In addition to chemical analyses, physical and geotechnical analyses of samples for characteristics, such as grain size distribution, will be required. Microscopic examination of building materials may also be required for the detection of asbestos and lead in paint samples.

Analyses, preferred methods, and reporting limits are task order specific. USACE-LRD maintains the ability to request analyses be conducted that are not specifically listed in Schedule A (Attachment 1) with the ability to negotiate per task order such as PFAS and fish tissue testing.

In summary, projects may consist, in part or total, of the following:

- Chemical, physical, and biological analysis of surface water, groundwater, sediments, soil, and tissue samples.

- Predictive chemical release testing via elutriate and leaching tests according to procedures described herein. Schedule A (Attachment 1) includes a list of generally anticipated analytical requirements for individual task orders. Other specialized or unique parameters required for analysis for any given individual task orders required under the above State or Federal compliance programs will be negotiated by task order, including the use of any implements for sample collection such as Terracores for VOC soil and sediment analysis.

2.4 Objectives

The objective of work specified through this contract is to obtain technically valid and legally defensible environmental data that will meet or exceed the site-specific Data Quality Objectives (DQOs) as identified in each task order.

2.5 Period of Performance

Period of Performance (PoP) of services will be from the time of receipt of the contract for five (5) years. PoP for each task order shall be determined individually on a task order basis.

2.6 Request for Proposals

The Contracting Officer will send a request for proposal (RFP) with a Task Order PWS._ Negotiations, if necessary, will be scheduled after review of the Contractor's proposal.

2.7 Points of Contact

The Contractor shall designate an individual to serve as the point of contact for the USACE on any contract and task order correspondence. USACE points of contact are the Contracting Officer's Representative (COR) and the Project Coordinator. See Section 5.1 "Key Personnel" for further information.

3 SERVICES REQUIRED

3.1 Sample Handling

3.1.1 Sample Containers

The Contractor shall provide the appropriate quantity of sample containers with the required preservatives and additional packaging (i.e. bubble-wrap envelopes) for all requested analyses. Sufficient coolers shall be provided by the laboratory for transport of all samples to and from the analytical laboratory after sampling is complete, including allowance for preservation and packing materials required during shipping. Sample containers and coolers will be delivered to the respective District office, or any Field Office associated with the LRD District requesting services, or the project site as indicated in each task order. The Contractor shall provide the FedEx account number (or account number associated with any overnight courier preferred by the Prime Laboratory) associated with the Prime Laboratory to ensure overnight shipment of collected samples.

3.1.2 Laboratory Sample Management Sample Handling Facilities

The Contractor shall provide an adequate, contamination-free, and well-ventilated workspace for the receipt of samples. All samples and their associated extracts must be stored under conditions that will preserve their integrity. Samples to undergo physical analysis, such as grain size determination, or samples for biological analyses, must not be frozen, compressed, drained, or dried. Sufficient refrigerator space must be provided for the proper storage of all appropriate samples and their extracts for a minimum of six months after receipt of the final data report. After that time, the laboratory is responsible for the disposal of the samples in compliance with all federal, state, and local regulations at the laboratory's expense.

3.1.3 Laboratory Custody Procedures

Chain-of-custody procedures shall be maintained in the laboratory from the time of sample receipt to the time the sample is discarded. The following procedures shall be followed in the laboratory. All incoming samples shall be received by a Sample Custodian, who shall indicate receipt by signing the accompanying custody forms and shall see that the signed forms are maintained in the project file as permanent records. Individual cooler receipt forms will be used for each shipment to verify and document any problems noted. The Contractor's Sample Custodian shall be assigned for each task order.

The Data Manager shall be notified of receipt and shall post chain-of-custody/analysis assignment sheets. The Data Manager shall also maintain a permanent bound logbook to record, for each sample, information such as location number, date sampled, lab/lot number. The Data Manager and Sample Custodian shall ensure that samples are properly stored and maintained until disposal instructions are received from the Laboratory Manager. All calculations and results shall be recorded in laboratory notebooks by the analyst at the time of analysis except for computer generated calculations. The Contractors Sample Custodian, Data Manager, and Laboratory Manager shall be assigned for each task order. Key Personnel and Points of Contact shall be identified in the Quality Control Plan.

3.2 Sample Analysis

3.2.1 Analytical Techniques

The methods used for preparing and analyzing water, soil, sediment, and biological samples are given in the following publications. The most current versions of these guidelines and regulations as of 2026 supersede any out-of-date or obsolete versions.

Reference 1: "Guidelines Establishing Test Procedures for the Analysis of Pollutants" Code of Federal Regulations, Title 40 (1995), Part 136.1 – 136.7. https://www.ecfr.gov/current/title-40/chapter-I/subchapter-D/part-136

Reference 2: Test Methods for Evaluating Solid Waste: Physical/Chemical Methods Compendium (SW-846). https://www.epa.gov/hw-sw846

Reference 3: American Public Health Association, American Water Works Association, and Water Environment Federation. 2023. "Standard Methods for the Examination of Water and Wastewater, 24th Ed.", Washington, D.C.

Reference 4: U.S. Army Corps of Engineers. 1981 (May). "Procedures for Handling and Chemical Analysis of Sediment and Water Samples," EPA/CE-81-1, Environmental Laboratory, U.S. Army Engineer Waterways Experiment Station, Vicksburg, Miss.

Reference 5: U.S. Environmental Protection Agency/U.S. Army Corps of Engineers. 1998 (September). "Great Lakes Dredged Material Testing and Evaluation Manual."

Reference 6: U.S. Environmental Protection Agency/U.S. Army Corps of Engineers. 1998 (February). "Evaluation of Dredged Materials Proposed for Discharge in Waters of the U.S. - Testing Manual, Inland Testing Manual".

Reference 7: U.S. Environmental Protection Agency/U.S. Army Corps of Engineers. 2003 (January). "Evaluation of Dredged Material Proposed for Disposal at Island, Nearshore, or Upland Confined Disposal Facilities – Testing Manual".

Reference 8: IEPA Material for Analysis for Dredge and Fill Activities. Section 401 Water Quality Certification.

Reference 9: U.S. Environmental Protection Agency/U.S. Army Corps of Engineers Technical Committee on Criteria for Dredged and Fill Material and Plumb, Russell H. 1981 (May). Procedures for Handling and Chemical Analysis of Sediment and Water Samples.

Reference 10: U.S. Environmental Protection Agency, Office of Water. 2000. "Guidance for Assessing Chemical Contaminant Data for Use in Fish Advisories. Volume 1. Fish Sampling and Analysis." EPA 823-R-93-002.

The Contractor shall ensure that samples are quantitatively analyzed using prescribed methods to concentration levels at least as low as the target quantitation limits provided in each task order DQOs. Sediment and soil constituent detection limits are reported on a dry weight basis. Due to stringent requirements to evaluate sediment and water samples in comparison to ambient concentrations in the sediments and waters of the Great Lakes region, the achievement of these limits may require the use of alternate methods or modification of the cited methods using larger sample sizes, smaller extract volumes, cleanup procedures, etc per task order. Any necessary method modifications shall be included in the Contractor's Quality Control Reports, as amended for each task order, and be approved by the USACE Project Coordinator. QC Reports must be included with each draft and final data submittal. The Contractor shall notify the USACE Project Coordinator immediately if the specified detection limit cannot be obtained using the recommended method. Equivalent sample preparation and analysis procedures which can achieve the target detection limits may be acceptable. Any equivalent test proposed for use shall be described in detail in task order proposals; statistical verification may be requested. The final report shall include a table citing the analytical method used for each analytical parameter.

Samples under this contract shall be stored as described below for six months following delivery of the final report to the USACE. Analytical results must be reported within thirty (30) days from the time samples are received unless otherwise specified in the task order.

3.2.2 Sample Extraction Procedures

3.2.2.1 Supernatant and/or Elutriate Preparations

A supernatant and/or elutriate test consists of mixing specific amounts of water and sediment to determine how much sediment constituent releases to the water column. Guidance must be followed as provided in References 5 through 9. The Schedule A price shall include the cost of conducting supernatant or elutriate procedure only. Note that resultant water samples would be analyzed for parameters and target detection limits specified in individual task orders.

3.2.2.2 Extraction/Leaching Procedures

The USACE may request other extraction procedures such as the ASTM Solid Waste Extraction test "Shake Leachate" (D3987-12), the Toxicity Characteristic Leaching Procedure (TCLP, EPA Method 1311), or the Synthetic Precipitation Leaching

Procedure (SPLP, EPA Method 1312). The Schedule A price shall include the cost of conducting the test and generating the extract only. Note that extract would be analyzed for parameters and target detection limits specified in individual task orders.

4 CERTIFICATIONS

The laboratory must maintain accredited status throughout the life of this contract. Loss of accredited status at any time during the term of this contract may result in termination of this contract.

4.1 State Certification

Laboratories must maintain National Environmental Laboratory Accreditation Program (NELAP) accreditation for all relevant fields-of-testing, which includes on-site evaluations, proficiency testing, and compliance with ISO/IEC 17025 standards. They must also obtain and maintain specific certifications required by the individual states where analyses are performed. Additional costs associated with obtaining or maintaining certifications are a Contractor expense. Copies of active certificates shall be provided to USACE upon contract award and within the Quality Control Plan for each individual task order.

4.2 USACE HTRW Compliance

Environmental laboratory services for HTRW projects, identified in individual task orders, are to be provided only by laboratories compliant with the most recently published version of the Department of Defense Quality Systems Manual (QSM) and holding a current NELAP accreditation for all appropriate fields-of-testing. This manual can be found at the following web site: http://www.navylabs.navy.mil/edqw/upload/QSM-Version-5-0-FINAL.pdf.

The Laboratory shall be accredited by the Department of Defense Environmental Laboratory Accreditation Program (DOD ELAP) for all applicable matrix and method combinations for HTRW projects. The certification documentation (including a copy of the DOD ELAP accreditation) will be submitted as a part of the bid package, prior to contract award.

5 QUALITY CONTROL AND QUALITY ASSURANCE

In all cases it shall be the responsibility of the Contractor to document that samples are analyzed using adequate quality control (QC) to assure the integrity of the analysis and to document the methods used in calculating results. Documentation should include conversion factors, dilution volumes, percent recovery, standard deviations, etc. For any analysis which the Contractor cannot provide adequate to describe the methodology used in achieving the results, and the accuracy of the results, the USACE may determine that the task order or an appropriate portion is unacceptable.

5.1 Key Personnel

The laboratory shall have a Quality Assurance (QA) Officer responsible for overseeing the quality assurance aspects of the data and reporting directly to upper management to meet all the terms and conditions of this contract. This individual shall have a minimum of a bachelor's degree in chemistry or any related scientific/engineering discipline. A minimum of three years of laboratory experience, including at least one year of applied experience with quality assurance principles and practices in an analytical laboratory, shall be required. The designated QA Officer will be responsible for assuring that data is within contract-specified limits and that transcription errors have been eliminated. The QA Officer's signature approving release will be required on all reports submitted under this Contract. Contractor functional area responsibilities to be designated specifically under this contract will include:

- Sample Custodian. The Sample Custodian's duties include:
    - Receives incoming samples, indicating receipt by signing accompanying custody forms
    - Ensure signed forms are maintained in the project file as permanent records.
    - Ensure proper storage and maintenance of samples until disposal instructions are received.

- QA Officer. The QA Officer's duties include:
    - Implementation of the QA/QC program during analytical activities.
    - Preparation and introduction of appropriate calibration standards, control sample standards, and blanks.
    - Review of QA/QC data.
    - Alerting the Laboratory Manager to out of control situations and committing resources to bring them back into control.

- Data Manager. The Data Manager's duties include:
    - Maintains a permanent, bound logbook to record sample number, sample data and laboratory number.
    - Records results in a computerized database.
    - Supplies analytical results to the QA/QC Manager.

USACE Key Personnel include both the Contracting Officer's Representative (COR) and the USACE Project Coordinator.

5.2 Quality Assurance and Quality Control Plans

This acquisition is for commercial services. The Contractor's existing quality assurance system shall be utilized in accordance with FAR 12.208.

Upon award of the contract, the contractor will develop and submit a Quality Control Plan (QCP) for review and acceptance by the USACE. The QCP will address QA/QC for submittals under this contract. The analytical laboratory shall submit the current Quality Assurance Plan (QAP). The QAP may be the commercial standard utilized by the laboratory. The elements of the Contractor's internal QA/QC program must consist of at least the following:

1) Documentation that analysis is performed by skilled and qualified personnel using approved, written, and validated methods.

2) Documentation of properly constructed, equipped, and maintained laboratory facilities.

3) Documentation of the use of high-quality glassware, reagents, and other testing materials.

4) Documentation of the calibration, adjustment, and maintenance of equipment.

5) Documentation of the use of control samples and standard samples with proper records.

6) Documentation of direct observation of the analyst's performance of certain critical tests by the QA Officer.

7) Documentation of the review and critique of results by supervisory personnel.

8) Documentation of the use of reference, replicate, and spiked samples for all matrices being analyzed.

9) Documentation of a reference sample program for interlaboratory comparisons, data comparability, and validity.

10) Documentation of the use of duplicates, spikes, or known quality control (reference) samples to verify accuracy of freshwater organisms’ collection, separation (when appropriate), and identification techniques.

11) Documentation of spot checks of laboratory personnel's organism identifications or analytical techniques by the laboratory supervisor or other qualified individuals.

12) Documentation of an adequate technical reference library including: Major journals and reference texts, appropriate EPA laboratory procedures manuals and appropriate taxonomic keys, and publications referenced under this document.

13) Documentation of response to customer complaints.

14) Documentation of correction of departures from standards of quality.

15) Documentation of a Laboratory Safety Program including exhaust hoods (flow checked), well-defined waste disposal policies/procedures, safety glasses, gloves, safety meetings, etc.

Each submittal of data (draft or final) must include an accompanying analytical Quality Control (QC) Report. An updated QCP should be provided per task order should the work requested include unique or specialized QC procedures, and an updated QAP should be provided per task order should it be updated by the laboratory compared to the base QAP.

5.3 Contractor Support Hours

The Contractor shall be available to provide support between the hours of *0900-1500, the days Monday through Friday, except during local or national emergencies, or administrative closings.* “Support” in this context includes electronic communication to manage each task order during its respective period of performance. The Contractor shall maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce is essential.

5.4 Laboratory Quality Control

5.4.1 Laboratory Documentation

In all cases it shall be the responsibility of the Contractor to document that samples were analyzed using adequate quality control (QC) to assure the integrity of the analysis and to document the methods used in calculating results. Documentation should include conversion factors, dilution volumes, percent recovery, standard deviations, etc. For any analysis which the Contractor cannot provide adequate documentation (to the satisfaction of the USACE) to describe the methodology used in achieving the results, and the accuracy of the results, the USACE may determine that the task order or an appropriate portion is unacceptable.

5.4.2 Laboratory Standards

As a part of the laboratory QC program, it is required that the Contractor analyze laboratory control samples (LCS) (for all matrices being analyzed) to verify accuracy of analytical system. Check standards shall be analyzed at specific intervals to verify the laboratory’s analytical process in operating within specifications. Calibration standards shall be used for instrument calibration and to establish control limits for analytical parameters. Matrix spike/matrix spike duplicate (MS/MSD) analysis of soil, sediment, and water samples shall be used to determine if there are interfering substances in samples that could affect reporting. As a rule of thumb, the QA/QC effort should constitute at least 15% of the analytical effort for each delivery order and matrix. Calibration and performance tests shall be routinely performed. Laboratory control samples and primary standards should be directly traceable to National Bureau of Standards or EPA Standards.

5.4.3 Data Quality Control

The specific role of various types of QC samples that may be analyzed are described in the following paragraphs. The number of field quality control samples required will be defined in individual task orders.

1) Trip Blanks. These blanks shall consist of duplicate, 40 ml vials, filled in the laboratory with organic-free DI water, transported to the site, handled like a sample without having been exposed to sampling procedures, and analyzed to ensure contamination is not present due to container preparation or shipping procedures. Typically, analyzed only for volatile compounds.
2) Field Blanks. These blanks shall consist of analyte free water poured into containers in the field, preserved and shipped to the laboratory with field samples. The Laboratory shall provide organic-free DI water for collection of these samples.
3) Rinsate Blank. Samples of analyte-free (deionized) water which are rinsed over decontaminated sampling equipment, collected, and submitted for analysis. These samples are used to assess cross-contamination from the sampling equipment, in addition to incidental contamination and/or the sample container.
4) Laboratory Blanks. These blanks shall be analyzed with each group of soil and water samples to identify those cases where laboratory procedures may be responsible for introduction of contaminants.
5) Split Samples. After collection, this sample is divided into two parts and sent to different laboratories for the same analyses. Soil samples shall be composited and thoroughly mixed before dividing.
6) Field Duplicates. These samples, collected at the same time and location and placed in separate sample containers, shall be used to assess the precision of the overall sampling and analysis procedures.

5.5 Laboratory Control Techniques

The Contractor shall demonstrate acceptable accuracy and precision by using QA/QC procedures that include analysis of spiked and spiked check samples. Routine sample analyses using each of the analytical methods to be used shall incorporate these procedures. The Contractor shall maintain performance records and control charts to determine whether results are within control limits for all of the analytical parameters. Controls for each surrogate parameter are defined by the equation:

$$\text{Control Limits} = R \pm 3s$$

Where R = Mean percent recovery for each surrogate
s = Standard deviation of percent recovery(ies).

Control of instrumentation and analytical procedures shall be maintained primarily through use of control limits established by analysis of samples, surrogates and standards. If the warning levels defined by $R \pm 2s$ is exceeded, potential sources of a problem such as sample preparation, calibration procedures, use of reagents, improper dilutions, calculation errors, or need for instrument maintenance shall be reviewed. If the control limits (R± 3s) are exceeded, the QA Manager shall further investigate the nature of the problem, and generally, recalibration and repetition of the analyses shall be required.

5.6 Precision, Accuracy, Representativeness, Comparability, and Completeness (PARCC) Parameters

The determination of data quality includes the evaluation of the PARCC parameters (i.e., precision, accuracy, representativeness, comparability, and completeness) and detection limits. The PARCC parameters represent quantitative limits below which data is unacceptable. The procedures described in this contract are designated to obtain PARCC data for each analytical method. To ensure that quality data is continuously produced, systematic checks must show that test results remain reproducible, and that the analytical method is measuring the quantity of target analytes in each sample. The reliability and credibility of analytical laboratory results can be corroborated by the inclusion of a program of scheduled analyses of replicates, standards, or spiked samples.

Regularly scheduled analyses of known duplicates, standards, and spiked samples are a routine aspect of data review, validation and reporting procedures.

1) Precision. Precision examines the distribution of the reported values about their mean. The distribution of reported values refers to how different the individual reported values are from the average reported value. Precision is a measure of the magnitude of errors which can be measured in a variety of ways. A simple measure of precision is the standard deviation. For chemical analysis of environmental samples, precision is commonly determined from duplicate samples; thus, precision is usually expressed as relative percent difference (RPD) or relative standard deviation (RSD). Every batch of samples analyzed shall include matrix duplicates and matrix spike duplicates to evaluate precision.

2) Accuracy. Accuracy measures the bias in a measurement system and is difficult to measure for the entire data collection activity. Sources of error are the sampling process, field contamination, preservation, handling, sample matrix, sample preparation, and analysis techniques. Sampling accuracy may be assessed by evaluating the results of rinsate and trip blanks. Analytical accuracy may be assessed using known and unknown QC samples and spiked samples. Accuracy values can be presented in a variety of ways. The average error is one way; however, accuracy is more commonly presented as percent bias or percent recovery. Percent bias is a standardized average error; that is, the average error divided by the actual or spiked concentration and converted to a percentage. Percent recovery provides the same information as percent bias. Accuracy is often determined from spiked samples. Every batch of samples analyzed shall include matrix spikes, laboratory control samples, and surrogate spikes, if appropriate.

3) Representativeness. Representativeness expresses the degree to which sample data accurately and precisely represent the characteristics of a population of samples, parameter variations at a sampling point, or an environmental condition. Representativeness is a qualitative parameter which is most concerned with the proper design of the sampling program or subsampling of a given sample. The representativeness criterion is best satisfied in the laboratory by making certain that all subsamples taken from a given sample are representative of the entire sample. The noting of sample characteristics in a case narrative will assist with the evaluation of the prime contractor's data. Representativeness can be assessed using duplicate field and laboratory samples. In this way, they provide both precision and representativeness information. Every batch of samples analyzed shall include matrix duplicates.

4) Completeness. Completeness is defined as the percentage of measurements made which are judged to be valid measurements. The completeness goal is essentially the same for all data uses in that a sufficient amount of valid data be generated. It is important that critical samples are identified, and valid data obtained. Due to the QA nature of the work to be performed under this contract, the highest degree of completeness that can be achieved is desired. The desired level of completeness is dependent on the specific site from where a group of samples may have originated. This information will be conveyed to the laboratory if it is known.

5) Comparability. Comparability is a qualitative parameter expressing the confidence with which one data set can be compared with another. Sample data should be comparable with other measurement data for similar samples and sample conditions. This goal is achieved through using standard techniques to collect and analyze representative samples and reporting analytical results in appropriate units. Comparability is limited by the other PARCC parameters because only when precision and accuracy are known can data sets be compared with confidence. Due to the nature of the QA work to be performed under this contract, it is imperative that contract- required methods are explicitly followed. Any deviations shall be approved in advance. SOPs for all the sub-contract laboratory's methods and procedures will be maintained on file at the Contractor's facility. Before any procedures can be changed, appropriate SOPs must be received and approved by the USACE Project Coordinator.

5.7 Quality Control Reports

The Contractor must provide a Quality Control Report with each draft and final analytical report associated with a task order. This report must include the following:

- The results of split, blank, and/or reference samples.
- The procedures used in interpretation of the reference sample data.
- The spiked results for all standard methods with flags on any samples outside QC limits, and the methods for deriving QC limits.
- A Case Narrative section describing any instances where deviations from

standard protocols occurred. In instances where contract targets were not met the narrative should describe the problem and what sample cleanup methods were employed. In instances where deviations from contract methods occurred, the narrative should describe when the deviation was noted, the USACE contact who authorized the deviation, and what corrective actions were taken.

6 DATA HANDLING AND REPORTING

6.1 Data Management Data Management System

The preferred technique for management of USACE samples is that samples be cataloged into the laboratory information management system (LIMS) by the sample custodian when they arrive. The LIMS is defined as a computer-based data storage, management and reporting system designed and programmed to meet the needs of the Contractor laboratory. After the sample is cataloged, the LIMS "tracks" each sample throughout its analytical journey, providing laboratory managers with updates and status reports. When the sample analysis is completed, the data shall be distributed to the laboratory manager. The raw data sheets generated by the laboratory technicians and chemists shall be stored in bound notebooks. To ensure the quality of data entered into the LIMS, the data shall be verified by the QA Manager.

6.2 Data Reporting Format

Draft and final analytical reports shall be prepared and submitted as one complete report in adobe (pdf extension) form for review. The report title page shall identify the report title, "Prepared for the U.S. Army Corps of Engineers, USACE with the Task Order Number, project name, and date. The report shall include analytical results, case narrative, and the quality control/quality assurance report in accordance with the Contract.

Electronic data deliverable will be of a MS Excel format as .xls, .xlsx or .csv file, See Attachment 2. Analytical data entry requirements to be provided by the contractor include the project name, USACE Project Coordinator, sample identification, laboratory identification, sample matrix, latitude/longitude, collection/extraction and analysis dates, methods of analysis, batch number, CAS number, compound/element/isotope, sample result, total uncertainty, units, lab qualifier, detection limit, dilution factor, LCS calculated % recoveries with control limits, lab duplicate and MS/MSD calculated RPDs with control limits, a description of the sampling location, and the physical description of the sample.

Biological testing reports will include a case narrative describing the sample receipt and handling, a physical description of each composite sample, tests conducted, the results of each test (including results for the controls and replicates), any special directives provided by the USACE, any deviations from protocols, any analytical results requested for the composite samples, statistical analysis of the results, and the quality control report in accordance with the contract.

6.3 Report Submittals

The Contractor will submit documents as a draft final report and a final report; the Contractor shall ensure that all data packages are standard Level II. Other Levels may be negotiated as required per task order.

The final submittal shall be complete and incorporate all government comments made during the review period and shall consist of an electronic copy of the entire report in searchable adobe (pdf extension) format. Reports shall include the following items in the listed order, as applicable:

1. A title page with the project name and task order number.
2. Sediment/Soil Analytical data
3. Elutriate Analytical data
4. Water Analytical data
5. Biological survey results
6. QC report.

QC data should include sample number, sample identification, and characteristic information (or adequate cross reference to this information), analysis parameter, parameter value, concentration units, detection limits, special problems encountered during analysis, and date the sample was received, etc.

The Contractor shall submit draft lab results per the submittal schedule.

The Contractor shall submit a draft final report for review by the Government. The report shall contain a title page that identifies the report title, “Prepared for the U.S. Army Corps of Engineers, Chicago District” with the Task Order Number and date. The draft final report shall include a case narrative describing the lab results.

The Contractor will submit a final report that incorporates changes resulting from Government review of the draft final report. The final report submittals will include the report in electronic format such as MS Word, MS Excel and/or GIS files. In addition, a single full report in searchable Adobe (in pdf format) should be created containing all the above-mentioned files. All files should be provided to the USACE COR.

6.4 Delivery

Written reports of services shall be furnished by the Contractor to the Government within the time frame identified in the statement of work for each task order. If a situation arises in which it becomes necessary to request an extension, i.e., if an order requires a longer period to complete, this request shall be made to the USACE Project Coordinator.

Completed reports will be forwarded to:

U.S. Army Engineer District, Chicago District
231 S. Lasalle Street, Suite 1500
Chicago, Illinois 60604
ATTN: USACE Project Coordinator (defined per task order)

6.5 Acceptance

The final submittal shall be provided within seven days of government acceptance or government comments on the draft report. If no comments are provided within 30 days of submittal, the draft report shall be deemed to be accepted and the final report shall be provided.

Required analytical data will be specified in the task orders by parameter name, matrix, and price. The test results may be accepted, and payment made from analytical items on a line-by-line, sample-by-sample basis if warranted after technical review. An extensive pattern of implausible data or QA/QC failures may result in a determination that the entire data set is unacceptable.

Final acceptance of the services by the Government will be considered as the time when each task order has been completed, and all terms and conditions of the contract have been met. The Performance Requirements Summary (PRS) is given in Attachment 3 and defines the Government's expectation of Contractor deliverables. Payment shall not be rendered until the data is reviewed for completeness and task order Scope of Work adherence, at a minimum.

7 PRICING INFORMATION

7.1 Services

7.1.1 Unit Prices

The unit prices for each parameter listed in Schedule A are firm fixed prices (whether an order calls for 1 test or as many as 50 or more). The unit price includes:

- Sample Digestion
- Sample Bottle Preparation
- Sample Preparation
- Sample Clean-up
- Quality Control Sample Preparation and Analysis
- Data reporting

Prices are all-inclusive.

7.1.2 Quality Control Measures

The normal QC routine shall include analysis of calibration standards, a travel blank or rinsate blank, and a laboratory control sample. Costs of these routine QC efforts should be included within the unit price of each parameter.

7.1.3 Analytical Methods

Recommended analytical methods will be defined for each task order. The list of parameters should be limited to the priority pollutants, unless otherwise specified in the task order.
Requests for PAHs would involve quantifying the PAH compounds listed below.

Acenapthene	Dibenzo(a,h)anthracene
Acenapthylene	Fluoranthene
Anthracene	Fluorene
Benzo(a)anthracene	Indeno(1,2,3-cd)pyrene
Benzo(a)pyrene	Naphthalene
Benzo(b)fluoranthene	Phenanthrene
Benzo(g,h,i)perylene	Pyrene
Benzo(k)fluoranthene	2-Methylnaphthalene
Chrysene	

8 DELIVERABLES

Most submittals are *per task order* unless otherwise annotated. The following list does not include deliverables that are required at solicitation and award (QAP, QCP). The deliverables schedule (typical schedule given in Attachment 4) and PoP will be specified per task order.

- Draft Lab Results (Section 6.4)
- Draft Final Reports (Section 6.4)
- Final Reports (Section 6.4)

9 PAYMENT REQUESTS

9.1 Invoices

Minimum requirements are given in FAR 52.212-4(g). Payment requests or invoices shall be submitted and contain the following information as a minimum: Project Title, dates covered for each invoice, total contract amount with all modifications and amounts listed individually. The invoice must be accompanied by supporting documentation for each task worked on during the billing period. The invoice must be a final invoice of all services completed; individual work items may not be separately billed. The COR cannot sign off on any invoice for which supporting documentation is not provided. When submitting invoices to the USACE Finance Center, the contractor shall promptly E-mail payments requests to the COR.

Payment shall not be rendered until the data is reviewed for completeness and task order Scope of Work adherence, at a minimum, during the draft review phase.

10 SECURITY REQUIREMENTS

10.1 General Security Requirements and Guidance

The security requirements described below apply to all contract personnel (including employees of the prime Contractor ("Contractor") and all subcontractor employees) supporting the performance requirements of this contract. The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed)). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training contributes to those efforts. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared to provide copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as "RAMs"), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures --this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

10.2 Suspicious Activity Reporting training (e.g. iWATCH, CorpsWatch, or See Something, Say Something)

All contract personnel shall receive initial and annual refresher training from the RA representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative or law enforcement entity (see Attachment 5).

10.3 Pre-screen Candidates Using E-Verify Program

Contractors shall comply with the requirements set forth in FAR clause 52.222-54. Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (https://www.e-verify.gov/) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

In addition to complying with the requirements outlined in FAR Part 22.13, FAR Provision 52.222-38, FAR Clause 52.222-35, FAR Clause 52.222-37, DFARS 222.13 and Department of Labor regulations, (USACE) contractors and subcontractors at all tiers are encouraged to promote the training and employment of U.S. veterans while performing under a USACE contract. While no set-aside, evaluation preference, or incentive applies to the solicitation or performance under the resultant contract, USACE contractors are encouraged to seek highly qualified veterans to perform services under this contract. The following resources are available to assist USACE contractors in their outreach efforts:

- U.S. Department of Labor Veterans’ Employment and Training Service (VETS): https://www.dol.gov/vets/
- Federal Veteran Employment Information: https://www.fedshirevets.gov/
- Veterans Opportunity to Work (VOW) Program: https://www.benefits.va.gov/vow/
- U.S. Army Warrior Transition Command Employment Index: https://wct.army.mil/modules/employers/index.html
- Hiring Our Heroes: https://www.uschamberfoundation.org/hiring-our-heroes

11 PUBLIC AFFAIRS

The Contractor shall not make available to news media or publicly disclose any data generated or reviewed under this PWS. When approached by the news media, public officials, etc., the Contractor shall refer them to:

U.S. Army Engineer District, Chicago
ATTN: Public Affairs Office
231 S. LaSalle Street, Suite 1500
Chicago, IL 60604
Phone: (312) 846-5553

Reports and data generated under this PWS shall become the property of the Government and distribution to any other entity by the contractor is prohibited, unless authorized by the USACE Project Coordinator.

ATTACHMENT 1 – Schedule A

Pricing Schedule - Environmental Sampling/Testing - Laboratory Rates								
ITEM #	DESCRIPTION	Matrix	Method(s)	UNIT PRICE	UNIT PRICE	UNIT PRICE	UNIT PRICE	UNIT PRICE
				Year 1	Year 2	Year 3	Year 4	Year 5
BIOLOGICAL SERVICES								
1	Benthic macroinvertebrate	Biota	Identification and counting (to the lowest practical taxonomic designation)					
2	Phytoplankton	Water	APHA Method A10200F					
3	Zooplankton	Water	Utermohl, oth					
4	Fecal Coliform, total	Water	9223B					
5	Total Coliform	Water	9223B					
6	Chlorophyll A,B,C and Pheophytin	Water	445/446					
CHEMICAL TESTING SERVICES								
7	Standard Elutriate preparation (SET)	Sediment/Soil	EPA/USACE. 1998. Evaluation of Dredged Material Proposed for Discharge in Waters of the U.S. – Testing Manual EPA-823-B-98-004 [ITM]					
8	Effluent Elutriate preparation (EET)	Sediment/Soil	USACE. 2003. Evaluation of Dredged Material Proposed for Disposal at Island, Nearshore, or Upland Confided Disposal Facilities – Testing Manual. Technical Report ERDC/ES TR-03-1. [UTM]					
9	Supernatant Elutriate preparation	Sediment/Soil	IEPA Material for Analysis for Dredge and Fill Activities. Section 401 Water Quality Certification.					
10	Shake Extraction of Solid Waste with Water	Soil/Sediment	ASTM D3987-85					
11	TCLP, Extraction Only (Volatile and non-Volatile portions)	Sediment/Soil	1311					
12	SPLP, Extraction Only (Volatile and non-Volatile portions)	Sediment/Soil	1312					
13	Reactivity	Sediment/Soil/Water	9012/9034					
14	Corrosivity	Sediment/Soil/Water	9045					
15	Ignitability/Flash Point	Sediment/Soil/Water	1010/1030					
16	Paint Filter Test	Sediment/Soil/Water	9095					
17	pH	Sediment/Soil/Water	9040/9045					
18	Metals - 23 metals per TAL	Sediment/Soil/Water	6010/6020/7471					
19	Metals - 8 metals per RCRA	Sediment/Soil/Water	6010/6020/7471					
20	Individual Metals, total	Sediment/Soil/Water	200.7/200.8/6010					
21	Individual Metals, dissolved	Water	200.7/200.8/6010					
22	Chromium Hexavalent - Cr + 6	Solid	7196/3060A					
23	Chromium Hexavalent - Cr + 6	Liquid	3500-CrB/7196					

24	Total Mercury	Sediment/Soil/Water	245.1/7470/7471					
25	Low Level Mercury, ng/L	Water	1631E					
26	Cyanide, Total	Sediment/Soil/Water	4500-CH-E/9012					
27	Volatile Organic Compounds – TCL	Sediment/Soil/Water	8260D					
28	BTEX (Benzene, Toluene, Ethylbenzene, Xylenes)	Sediment/Soil/Water	8260D					
29	Semi-Volatile Organics - TCL	Sediment/Soil/Water	625/8270					
30	Phenols	Sediment/Soil/Water	420.1/9065					
31	Polynuclear Aromatic Hydrocarbons – PAH	Sediment/Soil/Water	8270SIM					
32	PCB, total	Sediment/Soil/Water	608/8082					
33	PCBs, aroclors	Sediment/Soil/Water	608/8082					
34	Organochlorine Pesticides	Sediment/Soil/Water	608/8081					
35	Organophosphorus Pesticides	Sediment/Soil/Water	8141					
36	Chlorinated Hydrocarbons	Sediment/Soil/Water	8081					
37	Chlorinated Herbicides	Sediment/Soil/Water	8151A					
38	Oil and Grease	Sediment/Soil/Water	1664					
39	Common Anions (Cl^-, Br^-, F^-,NO3-, NO2-, PO43-, SO42)	Sediment/Soil/Water	9056/300.0					
40	Common Cations (Ca, Mg, Na, K)	Sediment/Soil/Water	200.7/6010C					
41	Chemical Oxygen Demand	Sediment/Soil/Water	410.4					
42	Biochemical Oxygen Demand (5 day)	Sediment/Soil/Water	405.1/A5210B					
43	Cyanide, total or free	Sediment/Soil/Water	4500/9012					
44	Nitrogen - Total Kjeldahl	Sediment/Soil/Water	351.2/ 4500					
45	Nitrogen - Ammonia	Sediment/Soil/Water	350.1					
46	Nitrogen - Nitrate	Sediment/Soil/Water	9056/300.0					
47	Nitrogen - Nitrite	Sediment/Soil/Water	353.2/354.1/300.0					
48	Sulfate	Sediment/Soil/Water	300					
49	Phosphorus, total	Sediment/Soil/Water	4500-P-F/4500 PH/365.4					
50	Soluble Reactive Phosphorus (SRP)	Sediment/Soil/Water	365.3/300.0					
51	Total Organic Carbon	Sediment/Soil/Water	415.1/9060					
52	Volatile Solids (%)	Sediment/Soil	2540G					
53	Total Solids (%)	Sediment/Soil	2540G					
54	Dissolved Solids (TDS)	Water	160.1/2540C					
55	Suspended Solids (TSS)	Water	150.2/2540D					
56	Total Volatile Solids (TVS)	Water	160.4					
57	Conductivity	Water	120.1/ 2510B					
58	Alkalinity	Water	2320B					
59	Hardness	Water	2340B					
60	Soluble Salts (Chloride and Sulfate)	Water	300					
GEOTECHNICAL TESTING SERVICES								
61	Mechanical Sieve Analysis	Solids	ASTM D6913					
62	Hydrometer Analysis	Solids	ASTM D7928					
63	USCS Classification	Solids	ASTM D2487					
64	Total Asbestos Fibers - Bulk	Solids	600/R-93/116					

ATTACHMENT 2 – NOAA Query Manager Format
NOAA DELIVERABLE FORMAT

NOAA (National Oceanic and Atmospheric Administration) - Laboratory Electronic Deliverable Format

Laboratory analytical results for the events are stored in a relational database management system. The objective in acquiring laboratory data as electronic data deliverables (EDDs) is to minimize the manipulation required to incorporate the data into the relational database management system. Laboratories submitting electronic data should provide either Excel (.xlsx) or ASCII tab delimited text files. Electronic files are due at the same time as the hard copy chemical reports. Table 1 below describes the fields required in the EDDs. The column labeled "Required?" indicates where the field must be completed.

A "Yes" indicates that the field must be completed for ALL records.

Table 1. Description of fields to be completed for laboratory electronic data deliverables (EDDs)

Field Name	Required?	Description
Project name	Yes	Name given to the project as designated on the chain of custody: e.g., Kalamazoo River
Sample delivery group	Yes	Identifier for the sample delivery group or quality control batch
Client sample ID	Yes, for client samples and some QC samples	Identifier as provided on the sample label. For matrix QC, (MS and lab duplicates) this should be the parent sample ID (without MS or D suffix). For non-project matrix QC (i.e., MS or LD performed on another client's samples), "XXXXX" or other identifier may be used. For laboratory generated samples (i.e. blanks and LCS), field may be left blank.
Matrix	Yes	Sample matrix such as water, sediment, tissue, oil, tarball, etc.
Laboratory sample ID	Yes	Identifier as used by the laboratory. A unique identifier for each analysis date and time is preferred.
Date collected	Yes	Date the sample was collected. The date should be formatted as mm/dd/yyyy. Where the month and/or day is a value less than 10, a leading zero should be used (e.g., 06/09/2001). Where the sample is lab generated (e.g., method blank, etc.), use "NA" for the date collected.

Field Name	Required?	Description
Date extracted	Yes	Date the sample was prepared or extracted in the laboratory. The date should be formatted as mm/dd/yyyy. Where the month and/or day is a value less than 10, a leading zero should be used (e.g., 06/09/2001). Where the sample is not extracted/prepared (e.g., Volatiles, etc.), use “NA” for the date extracted.
Date analyzed	Yes	Date the sample was analyzed. The date should be formatted as mm/dd/yyyy. Where the month and/or day is a value less than 10, a leading zero should be used (e.g., 06/09/2001).
Analytical method	Yes	Analytical method including laboratory SOP number, version and general instrumentation.
Preparation method	Yes	Preparation method including laboratory SOP number, version and general description.
CAS number	Optional	CAS registry number
Analyte name	Yes	Full chemical name
Result	Yes	Measured result or Method detection limit if not detected
Detection limit	Yes	Method detection limit (MDL) adjusted for dilution and varying sample sizes.
Reporting limit	Yes	Quantitation limit based on low standard adjusted for dilution and varying sample sizes.
Detect flag	Yes	Y for detections and N for non-detects
Laboratory qualifiers	Yes, but nulls allowed	Qualifiers associated with the result. The laboratory should provide definitions of all qualifiers used in supplementary documentation such as additional text file or additional worksheet in an Excel file.
Analyte Type	Yes	TRG for target analytes, SUR for surrogates, SPK for spike compounds, IS for internal standards and TIC for tentatively identified compounds.
Dilution factor	Yes	1 for undiluted samples; dilution factor for other results
Units	Yes	Units used for the concentration (e.g., mg/kg, mg/L).
Total or dissolved	Yes, for aqueous samples	For aqueous samples, indicate “tot” for total results and “dis” for dissolved fractions.
Measurement basis	Yes	Measurement basis for the result. Should be “dry” or “wet”.
Sample type	Yes	Indicator of sample type. See Table 2 for a list of appropriate sample types.

Field Name	Required?	Description
Instrument ID	Optional	A unique identifier specifying the Instrument the analysis was performed on.
Column ID	Optional	An identifier specifying the column used (primarily for GC methods)
SpikeConc	Yes, where applicable for QC samples	For spiked samples and surrogates, the concentration of the analyte added. For SRMs/CRMs or standards, the true value.
Percent Recovery	Yes, where applicable for QC samples	For spike samples and surrogates, the percentage recovery achieved.
RPD	Yes, where applicable for QC samples	For duplicates the relative percent difference. For multiple replicates, the relative standard deviation.
Lab name	Yes	Laboratory name or suitable short form name.

Table 2. Codes to use in Sample Type Field of EDD

Sample Type Code	Description
Samples and standard QC	
SMP	Sample submitted by field team
LD	Laboratory duplicate result (add number if there is more than one duplicate result e.g., LD, LD2, etc.)
MS	Matrix spike
MSD	Matrix spike duplicate
SRM	Standard reference material
MB	Method blank or preparation blank
LCS	Laboratory control sample
LCD	Laboratory control sample duplicate
Calibration and instrument QC (if supplied)	
CAL	Calibration standards (initial, continuing or other)
CB	Calibration or instrument blanks
PS	Post Spike
SRD	Serial Dilution
ICS	Interference Check Sample

Additional information concerning the format and content of the electronic data deliverables is provided below:

1. Each row of data should contain only one analyte for a given sample. Therefore, one complete sample will require multiple rows.
2. Each row should contain the following information as a minimum: date of sample

analysis, laboratory method, analyte name, measured result, reporting limit, detection limit, laboratory qualifiers, units, measurement basis (wet or dry), sample type, sample delivery group (SDG) or quality control batch, and lab name. All samples submitted by the client must also include the client sample ID.

3. If a spreadsheet program is used to produce the electronic deliverable, the values representing the measured concentration and reporting limit should be limited to the appropriate number of significant figures (i.e, there should be no trailing digits to the right of the decimal that express greater precision than is appropriate). Number formats used in Excel should not hide digits.
4. Laboratory samples for quality assurance/quality control should be clearly identified in the file (using the sample type designation as described above) and each should have a unique laboratory sample identifier.
5. If replicate analyses are conducted on a submitted sample, the laboratory sample identifier must be different on each of the replicates. The original analysis will be designated in the sample type field as “SMP”. The second analysis will be designated as “LD” in the sample type field. If additional analyses of the sample are completed, a number should be appended to the end of the “LD” code (e.g., LD, LD2, LD3, etc.).
6. An additional text file or worksheet in an Excel file should be provided with supplementary explanations for the data provided. This file or worksheet should contain the following information:
    - Name of laboratory
    - Sample delivery group (SDG) or batch number
    - Date on transmittal letter or case narrative of the hard copy report
    - Name and number of laboratory contact
    - Tabular summary of qualifiers used in the EDD and their definitions
    - A narrative of any missing data or limitations to the use or interpretation of the data
    - A list of any new sample type codes or analyte types that were used and their definitions.

ATTACHMENT 3 – Performance Requirements Summary (PRS)

Performance Objective	**Standard**	**Performance Threshold Acceptable Quality Levels (AQLs)**	**Method of Surveillance**
Sample Handling: *The Contractor shall provide the required containers and implements for sample transportation, ensuring proper custody documentation measures are followed.*	Section 3.1 and subsections	100% (acceptable): Zero deviation from standard. 0% (unacceptable): Multiple shipments of containers required after inspection reveals insufficient materials.	Coordination with Sample Custodian ahead of container shipment, thorough inspection of items shipped, thorough inspection of CoCs and receipts upon delivery.
Sample Analysis: *The Contractor will perform analyses on deposited samples consistent with current methods. Comments from USACE are incorporated into drafts, and data is amended, or analyses are re-performed as requested.*	Section 3.2 and subsections	100% (acceptable): Zero deviation from standard. 0% (unacceptable): Inspection of data reveals that data is unacceptable due to excessive flags, improper methods, etc. without communication with USACE to correct issues that arise in the draft submittals.	Data Quality Analyses performed by USACE.
Certification: *The Contractor will maintain relevant certifications for the methods required, in the State where analysis takes place.*	Section 4.0 and subsections	100% (acceptable): Zero deviation from standard. 0% (unacceptable): Requirements not met.	COR inspection during solicitation.
Quality Assurance and Quality Control: *The Contractor designates Key Personnel responsible for maintaining the QA/QC*	Section 5.0 through Section 5.7 and subsections	100% (acceptable): Zero deviation from standard, or deviations are reported with sufficient time for	Random Inspection of QAP/QCP.

plan, providing relevant laboratory documentation, upkeeping laboratory standards, and reporting deviations where necessary.		correction. 0% (unacceptable): QA/QC activities are not upkept, and deviations are reported without sufficient time to correct the error.	
Data Handling and Reporting: *The Contractor uses the preferred technique for data management, reports data as directed, provides requested submittals.*	Section 6.0 through Section 6.5 and subsections	100% (acceptable): Zero deviation from standard, or communication regarding delays is consistent and established early. 0% (unacceptable): Data is provided late without notice.	Data packages at all levels of completeness.
Deliverables provided on Schedule	Section 8.0	100% (acceptable): Zero deviation from standard or communication regarding delays is consistent and established early. 0% (unacceptable): Deliverables are provided late without notice.	Dates of delivery.

ATTACHMENT 4 – Typical Deliverables Schedule

Deliverable/Task	Timeframe
Final Commercial Quality Control Plan*	Submitted with proposal
Final Quality Assurance Plan *	Submitted with proposal
Draft Lab results	30 calendar days from sampling event
Quality Control Reports	Submitted with Draft and Final Data Submittals
Government Review and Comment	Allow 10 Business days
Final Lab Report and Summary Table	Submitted 15 Business days from receipt of Government review comments
Government Acceptance of Final Lab Report, final payment, performance evaluations, etc.	Allow 30 calendar days from submission of Final Reports

All items may be electronically delivered to the COR and/or Project Coordinator as defined in the PWS.

*Submitted at solicitation/award of base Contract. An updated QCP should be provided per task order should the work requested include unique or specialized QC procedures, and an updated QAP should be provided per task order should it be updated by the laboratory compared to the base QAP.

ATTACHMENT 5 – iWatch Army Brochure